

Coupled hydraulic and dielectric material properties of soil¹

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ABSTRACT

In fine-grained soils the movement of water is influenced by different surface bonding forces due to interface processes. The matric potential is a measure of the bonding forces on water in the soil. Based on a thermodynamic relationship between soil matric potential and dielectric relaxation behavior of water in different binding states a broadband dielectric relaxation model was developed which considers low frequency dispersion to 1 MHz as well as losses due to direct current conductivity. The sensitivity of the model on soil suction was systematically analyzed based on a pedotransfer function for soil textures ranging from sand to clay.

1 INTRODUCTION

High frequency electromagnetic determination of moisture in porous media, e.g. soil, is based on the strong relationship between volumetric water content and relative dielectric permittivity. In particular, in fine-grained soils the movement of water is influenced by different surface bonding forces due to interface processes [1]. The interface effects lead to a number of dielectric relaxation processes (free and bound water phase, Maxwell-Wagner-effect (MWE), counterion relaxation effects (CRE), [2, 3, 4, 5]). These relaxation processes are the reason for the strong frequency dependence of the electromagnetic material properties of soil below 1 GHz [6, 7], especially for clay rich soils [4, 8, 9]. However, in most applications this is not sufficiently taken into account [10].

The electromagnetic properties of the bound water phase in soils are the result of the interaction between the aqueous pore solution and the solid grain due to a diffuse electrical double layer (EDL). The EDL is substantially influenced by the specific surface area of the soil, the chemical composition of the pore fluid and the mineralogical composition of the solid grains [11]. Particularly, in fine grained soils the bound water phase is the link between the hydraulic and dielectric soil properties [12]. The matric potential is a measure of the bonding forces between water and soil [1]. For these reasons, Hilhorst [13, 14, 12] suggests a novel approach based on the relationship between soil matric potential and dielectric relaxation behavior of free and bound water, respectively.

Up to now only few experimental investigations exist on the influence of soil suction on the electromagnetic material properties of unsaturated soils [15, ?]. Furthermore, the dependence of the permittivity of bound water on the distance from particle surface, with respect to salinity of free pore water, the linked MWE and CRE are not adequately considered in the approach of Hilhorst. Moreover, systematically theoretical or experimental investigations of the sensitivity and validity of this approach are not available. In this study based on the Hilhorst-approach

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a generalized dielectric relaxation model was developed. The sensitivity of the model on soil suction was systematically analyzed based on a pedotransfer function (PTF) suggested by [17]. The results are compared with frequently used empirical [18, 19] and theoretical mixing models [20].

2 DIELECTRIC RELAXATION BEHAVIOUR OF MOIST SOIL

Organic free soil as a typical porous medium consists of three phases: solid particles (various mineral phases), pore air and a pore fluid. In principle the fractions of the soil phases vary both in space (due to composition and density of the soil) and time (due to changes of water content, porosity, pore water chemistry and temperature). The electromagnetic properties of the solid particles are frequency independent in the considered temperature-pressure-frequency range. The real relative dielectric permittivity varies from 3 to 15 [21]. The pore fluid is mainly an aqueous solution with a temperature-pressure-frequency dependent relative complex dielectric permittivity according to the modified Debye model [22, 23]:

$$\varepsilon_w^*(\omega, T, p) - \varepsilon_\infty(T, p) = \frac{\Delta\varepsilon(T, p)}{1 + j\omega\tau_w(T, p)} - j\frac{\sigma_{DC}(T, p)}{\varepsilon_0\omega} \quad (1)$$

with angular frequency $\omega = 2\pi f$, imaginary unit $(-1)^2 = j$, permittivity of free space ε_0 , direct current conductivity σ_{DC} , high frequency limit of permittivity ε_∞ , relaxation strength $\Delta\varepsilon = \varepsilon_S + \varepsilon_\infty$ with static dielectric permittivity ε_S . The dielectric relaxation time τ_w depends on temperature T and pressure p according to the Eyring equation [3, 22]:

$$\tau_w(T, p) = \frac{h}{k_B T} \kappa \exp\left(\frac{\Delta G_w^\#(T, p)}{RT}\right). \quad (2)$$

h denotes the Planck-constant, k_B -Boltzmann constant, $\kappa \approx 1$ the transmission coefficient, R gas constant and $\Delta G_w^\#(T, p) = \Delta H_w^\#(T, p) - T\Delta S_w^\#(T, p)$ free enthalpy of activation with activation enthalpy $\Delta H_w^\#(T, p)$ and activation entropy $\Delta S_w^\#(T, p)$.

The water or matric potential of a solid or liquid sample can be found by relating the sample water potential reading to the vapor pressure of air in equilibrium with the sample. The relationship between the sample water potential Ψ_m (in J/m^3 or Pa) and the vapor pressure of the air is:

$$\Psi_m(T) = \frac{RT}{V} \ln(\varphi(T)) \quad (3)$$

where $\varphi = e/E$ is relative humidity, e the vapor pressure of the air, E the saturation vapor pressure at sample temperature, R the gas constant, T the absolute temperature of the sample, and V the partial molecular volume of water. Therefore the relationship between Ψ_m and free enthalpy of activation $\Delta G_w^\#$ of free water under atmospheric conditions is given by:

$$\Psi_m(T) \cdot V = \Delta G_w^\#(T) - \Delta G_d^\#(T) \quad (4)$$

Equation (4) can be used to calculate the free enthalpy of activation of bound water $\Delta G_d^\#(T)$ with the free enthalpy of activation of water at a reference state $\Delta G_w^\#(T)$ (10.4 kJ/mol at atmospheric conditions and $T=293.15$ K). For this reasons, Hilhorst et al. [13, 12, 14] states the following assumptions to model the dielectric relaxation behavior of soil from the matric potential:

- The relaxation behaviour of water (free and bound) is computed according to (1). The direct current conductivity is neglected.
- The relaxation parameters in (1), high frequency limit of permittivity ε_∞ , and static relative dielectric permittivity ε_S are independent of the soil suction.

Table 1: Soil hydraulic parameters^b as a function of soil texture (USDA-System) according to [17], data for a natural sand used for the construction of a dike model [24] as well as a silty clay loam from a flood protection levee of the river Unstrut in Thuringia/Germany (SUT) [10].

soil	θ_S	θ_r	k_f	α	n	texture [%]		
	[cm ³ /cm ³]	[cm ³ /cm ³]	[cm/h]	[cm ⁻¹]	[-]	C	SI	S
C	0.38	0.068	0.20	0.008	1.09	100		
CL	0.41	0.095	0.26	0.019	1.31	35	40	25
SI	0.46	0.034	0.25	0.016	1.37		100	
S	0.43	0.045	29.7	0.145	2.68			100
SL	0.41	0.065	4.42	0.075	1.89	12	28	60
SLM *	0.33	0.037	-	0.036	5.42			100
SUT $\square$	0.55	0.090	-	0.003	1.25	29.7	49.8	20

^b Van Genuchten parametrization [25]. * The parameters are obtained from the primary drainage curve with a density index $D_r = 83\%$. $\square$ The Van Genuchten parameters are obtained from grain size distribution.

- The complex dielectric permittivity of the soil is modeled by a theoretical mixing approach according to Hilhorst et al. 2000 [14].
- The activation entropy $\Delta S_w^\sharp$ is assumed to be independent of the binding forces and therefore $\Delta G_w^\sharp(T)$ is approximated with $\Delta H_w^\sharp(T)$.

In contrast to [14], in this study there will be no assumption of the activation entropy $\Delta S_w^\sharp$, and $\Delta G_i^\sharp(T, p)$ is used instead of $\Delta H_i^\sharp(T)$. The dielectric relaxation time τ_w , according to equation (2), at the vapor-water interface is calculated with the free activation enthalpy of a reference state $\Delta G_w^\sharp(T)$ and measured matric potential according to (4). This enables the calculation of the relaxation time distribution of water as a function of the appropriate bonding state in dependency of the water content. The complex relative dielectric permittivity of free and bound water in dependence of the volumetric water content θ at atmospheric conditions then becomes:

$$\varepsilon_T^*(\theta, T) = \int_{\Psi(0)}^{\Psi(\theta)} \varepsilon_w^*(\Psi(\theta), T) \frac{d\theta(\Psi)}{d\Psi} d\Psi. \quad (5)$$

In this form the direct current conductivity $\sigma_{DC}(T)$ is implicitly included regardless the mentioned assumptions. However, Hilhorst 1998 (HIL, [13]) suggests the following equation to calculate the effective complex relative dielectric permittivity of soil:

$$\varepsilon_{r,\text{eff}}^*(\theta, T) = \Omega_H(\theta, T) + (1 - n)\varepsilon_G(T) + (n - \theta). \quad (6)$$

with porosity n and permittivity of solid grain at 293.15 K $\varepsilon_G = 4.5$. For the calculation of the bonding state of water as well as considering the pore water conductivity we use the term $\Omega_H(\theta, T)$ in the following form ([13, ?]):

$$\Omega_H(\theta, T) = \frac{\varepsilon_S(T) - \varepsilon_\infty}{3(2n - \theta)} \cdot \int_{\Psi(0)}^{\Psi(\theta)} \left[1 + j \frac{\omega h}{k_b T} e^{\frac{\Delta G_w(T)^\sharp + |\Psi_m(\theta)|V}{RT}} \right]^{-1} \frac{d\theta(\Psi)}{d\Psi} d\Psi + \theta \left(\varepsilon_\infty - j \frac{\sigma_{DC}(T)}{3\omega\varepsilon_0(2n - \theta)} \right). \quad (7)$$

In addition to the HIL mixture rule the complex refractive index model (CRIM, [21]) is used:

$$\varepsilon_{r,\text{eff}}^{*a}(\theta, T) = \Omega_{CRIM}(\theta, T) + (1 - n)\varepsilon_G(T)^a + (n - \theta) \quad (8)$$

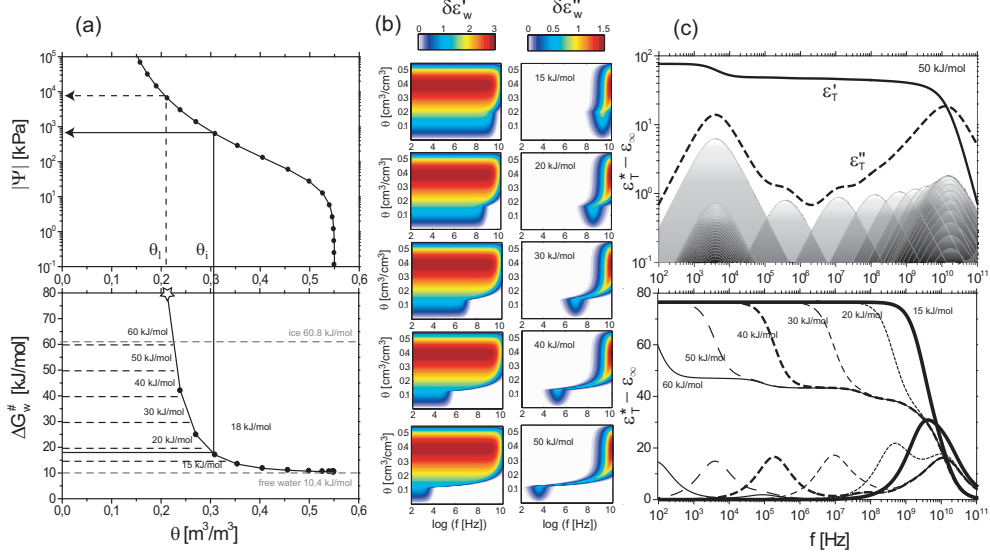


Fig. 1: (a, top) Soil water characteristic curve (SWCC, $|\Psi|(\theta)$), (a, bottom) free activation enthalpy $\Delta G_d^\#$ as a function of volumetric water content θ at 293.15 K. (b) Incremental real and imaginary part of the complex dielectric permittivity of water as a function of volumetric water content θ . (c, top) Total saturated relative complex dielectric permittivity ϵ_T^* of water according to (5) of the SUT-soil with an upper limit of the free enthalpy of activation $\Delta G_d^\# = 50$ kJ/mol (σ_{DC} is subtracted). (c, bottom) ϵ_T^* for different upper limits of $\Delta G_d^\#$.

with a so called structure parameter $a = 0.5$. The term $\Omega_H(\theta, T)$ then has the following form:

$$\Omega_{CRIM}(\theta, T) = \int_{\Psi(0)}^{\Psi(\theta)} \epsilon_w^{*0.5}(\theta, T) \frac{d\theta(\Psi)}{d\Psi} d\Psi. \quad (9)$$

3 MATERIAL

A parameter study was carried out to examine the suitability and the sensitivity of the Hilhorst approach relative to the appropriate soil suction. For this purpose the soil water characteristic curve (SWCC) were systematically varied based on the pedotransfer function (PTF) of [17]. In addition, a well-graded clay free sand of a levee model (SLM) [24] as well as a silty clay loam from a flood protection levee of the river Unstrut in Thuringia/Germany (SUT) [10]) with known SWCC is studied. The parameters of the van Genuchten-model as well as soil texture are summarized in Table 1. In Fig. 1 the contribution due to several relaxation processes as a consequence of (4) is represented. The upper limit of free enthalpy of activation determines the occurrence of the slowest relaxation process at the limit of the volumetric water content θ_l at which the calculated free enthalpy of activation $\Delta G_w^\#$ according to (4) diverges (see Fig. 1). This has pronounced consequences on the frequency dependence of the complex relative dielectric permittivity and therefore, on the modeled dependence on volumetric water content at a frequency of 1 GHz (Fig. 2). Thus, the dependence of the complex permittivity on volumetric water content is a measure of the relaxation time distribution of water in the different bonding states. As a fair assumption, an upper limit of $\Delta G_w^\# = 18$ kJ/mol is suggested. This is in accordance with [11] which predicts ≈ 18 kJ/mol for a distance of one molecule from the particle surface (see [?] for details).

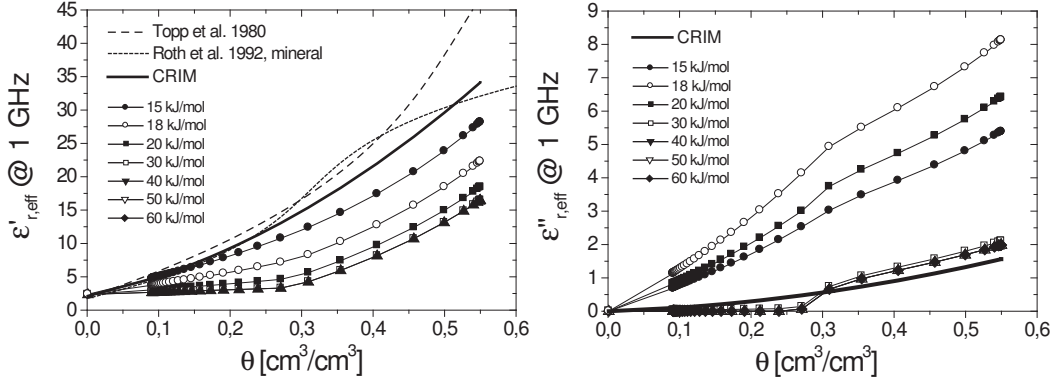


Fig. 2: Real part $\epsilon'_{r,\text{eff}}$ and imaginary part $\epsilon''_{r,\text{eff}}$ of the complex relative effective dielectric permittivity $\epsilon^*_{r,\text{eff}} = \epsilon'_{r,\text{eff}} - j\epsilon''_{r,\text{eff}}$ of the SUT-soil at a measurement frequency of 1 GHz and 293.15 K, as a function of volumetric water content θ for different upper limits of the free enthalpy of activation $\Delta G_w^\#$.

4 DISCUSSION

In Fig. 3 the results of the Hilhorst approach for five soils are represented. As expected the clay-soil (C) shows the highest and the sand (S) the lowest dielectric losses, respectively. The soils exhibit a markedly transition point or critical water content θ_C due to the bonding state of water. For the soils C, CL, SI θ_C corresponds to θ_i at a matric potential of 0.89 MPa due to the assumed upper limit of the free enthalpy of activation $\Delta G_d^\# = 18$ kJ/mol (see Fig. 1). For the soils SL, S and SLM θ_C corresponds with different matric potentials $\ll 0.89$ MPa. The specific matric potential is given by the SWCC of the soil. Thus, the measurement of the θ - ϵ relationship at a frequency of 1 GHz enables the determination of the upper limit of the free enthalpy of activation due to θ_C for a known SWCC.

The real part $\epsilon'_{r,\text{eff}}$ of the complex relative permittivity at a measurement frequency of 1 GHz deviates from the well known calibration function of Topp et al. 1980 [19] for all soils. Topp et al. 1980 determined the effective dielectric permittivity from TDR measurements with a 1 m and a 30 cm long coaxial probe. Therefore, a dependence of the effective frequency on moisture has to be expected. The effective measurement frequency f_{eff} which corresponds with an apparent real relative permittivity ϵ_{app} of the TDR-Signal is decreasing with increasing water content due to the low pass behavior of the soil, especially in soils with a marked clay fraction. This frequency shift effect the apparent relative permittivity considerably [10]. The Topp-model underestimates the volumetric water content of the soils with the SWCC parametrization, according to [17]. The clay soils (C, CL) show the largest difference. The frequently used empirical calibration function according to [18], determined from TDR-measurements with a 5.4 cm long two rod probe, shows a considerably smaller difference for the sandy soils (S, SI, SL) up to $0.25 \text{ cm}^3/\text{cm}^3$. In the experimental investigations of [18] f_{eff} is higher and the frequency shift is expect to be smaller than in the investigations of [19] due to the probe design and measurement equipment. However, both empirical models drastically overestimate the volumetric water content of the levee model soil (SLM) in comparison to the θ - ϵ relationship calculated with equation (6) (c.f. Fig. 3, see [?]).

In Fig. 4 the resulting frequency dependence of the complex effective dielectric permittivity of the SUT-soil calculated with both approaches (HIL, CRIM) is represented. $\epsilon'_{r,\text{eff}}$ in the frequency range above 1 MHz calculated with the CRIM-model is systematically higher than the HIL-model due to the strong influence of the porosity in the depolarization factor $S = 1/(3(2n - \theta))$. As mentioned in the introduction, soils with a significant clay fraction exhibit,

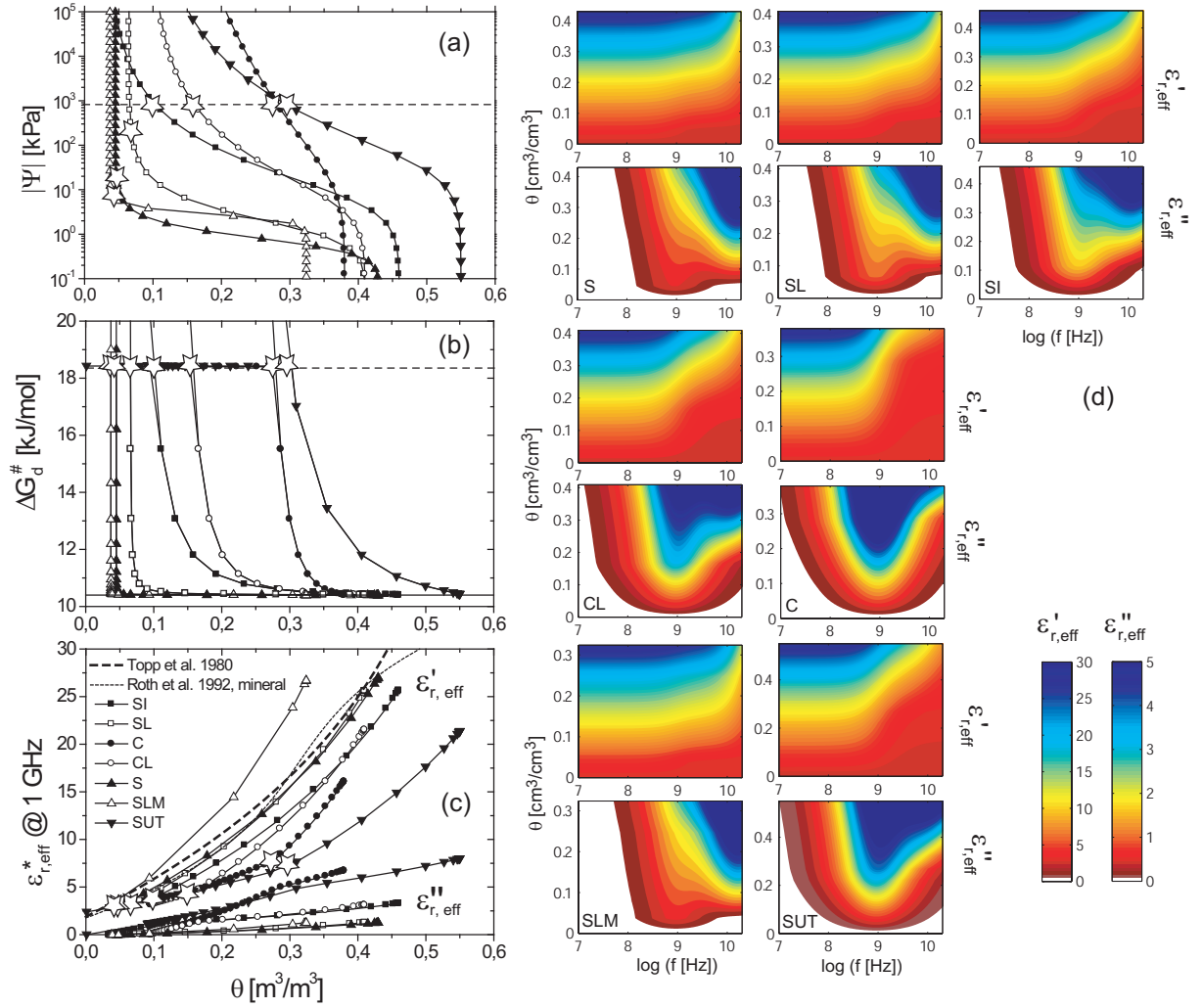


Fig. 3: (a) Matric potential $|\Psi|$, (b) free activation enthalpy $\Delta G_w^\ddagger$ and (c) complex relative dielectric permittivity $\varepsilon_{r,eff}^* = \varepsilon_{r,eff}' - j\varepsilon_{r,eff}''$ according to (6) at a frequency of 1 GHz and 293.15 K as a function of volumetric water content θ . Direct current pore water conductivity σ_{DC} is set to 0 S/m. Matric potential is calculated according to the van Genuchten parameters in Table 1. (d) Dielectric spectra as a function of volumetric water content for the appropriate material (dielectric fingerprint).

in the frequency range below ≈ 100 MHz, a strong frequency dependence of the real part of the relative dielectric permittivity due to MWE or CRE [6, 7, 3, 10]. With the HIL mixing rule this frequency dependence of the real part of the complex permittivity is not considered. The frequency dependence of the imaginary part is in accordance with observations of [10] for high volumetric water contents above $\approx 0.1 \text{ cm}^3/\text{cm}^3$ with a frequency factor $m = -\frac{\partial \log(\varepsilon_{r,eff}'')}{\partial \log \omega} = 1$. For a water content below $\approx 0.1 \text{ cm}^3/\text{cm}^3$ the influence of the direct current conductivity is expected to be much lower resulting in $m < 1$. In the CRIM-model the MWE interface polarization is considered and adequately predicts observations of [10] in the frequency range up to approximately 10 MHz. However, as in the case of the HIL model $m = 1$ for low water contents. Furthermore, especially in the case of a marked fraction of swelling clay minerals, i.e. smectite, CRE have to be considered. Equation (9) can be improved by adding an CRE relaxation term σ_{LF}^* . However, there is a lack of systematic investigations of the magnitude of the CRE in soils as a function of water content, porosity and chemical composition of the pore

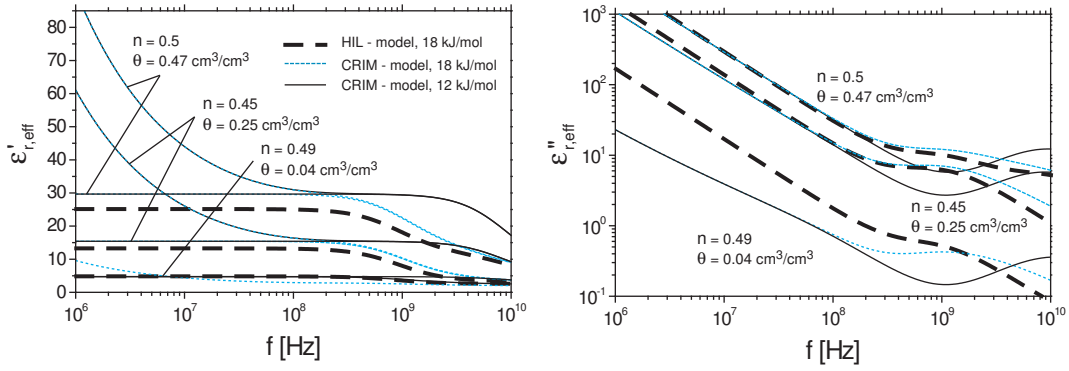


Fig. 4: Complex relative effective dielectric permittivity $\epsilon_{r,eff}^* = \epsilon'_{r,eff} - j\epsilon''_{r,eff}$ as a function of frequency for different volumetric water contents θ at a upper limit of free enthalpy of activation $\Delta G_d^\ddagger = 18 \text{ kJ/mol}$ and a direct current conductivity of pore water $\sigma_{DC} = 1 \text{ S/m}$.

fluid (c.f. [10]) therefore the CRE is not included in the model. The bound water relaxation occurs in the frequency range between 10 MHz and 100 MHz but is masked due to the MWE or CRE in high surface area soils.

5 CONCLUSION

A broadband generalized dielectric relaxation model was developed based on a thermodynamic approach for the link between soil matric potential and relaxation time of free and bound water suggested by Hilhorst [13, 14, 12]. With this approach, the influence of the physically bound water phase on the θ - ϵ relationship at a frequency of 1 GHz for fine grained soils is quantified, and the resulting differences to frequently used theoretical and empirical models are demonstrated. However, the Hilhorst-approach fails for soil with marked electrical losses, (e.g. with strong dispersion and absorption of the electromagnetic properties (MWE or CRE)), but the consideration of these effects in applications in the frequency range $< 1 \text{ GHz}$ is highly important [10]. The use of the CRIM model in combination with the Hilhorst-approach improves the model markedly. As expected the contribution of the bound water phase to the dielectric relaxation behaviour of the sand-soils in the frequency range above 1 MHz can be neglected. Nevertheless, for fine grained soils there exists an obvious combination of hydraulic and electric or dielectric material properties, respectively. The Hilhorst-approach offers the possibility to combine these properties via the SWCC. In the future, experiments have to be realized aiming at the quantitative investigation of the coupled hydraulic/dielectric properties of soils.

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